

RESEARCH ARTICLE

Stabilized multifunctional phase change materials based on carbonized Cu-coated melamine foam/reduced graphene oxide framework for multiple energy conversion and storage

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Abstract

The leakage of organic phase change materials (OPCMs) at temperatures above their melting point severely limits their large-scale application. The introduction of porous supports has been identified as an efficient leakage-proofing method. In this study, a novel carbonized Cu-coated melamine foam (MF)/reduced graphene oxide (rGO) framework (MF/rGO/Cu-C) is constructed as a support for fabricating stabilized multifunctional OPCMs. MF serves as the supporting material, while rGO and Cu act as functional reinforcements. As a thermal energy storage material, polyethylene glycol (PEG) is encapsulated into MF/rGO/Cu-C through a vacuum-assisted impregnation method to obtain PEG@MF/rGO/Cu-C composite with excellent comprehensive performance. PEG@MF/rGO/Cu-C exhibits high phase change enthalpies of 148.3 J g^{-1} (melting) and 143.9 J g^{-1} (crystallization), corresponding to a high energy storage capability of 92.7%. Simultaneously, MF/rGO/Cu-C endues the composite with an enhanced thermal conductivity of $0.4621 \text{ W m}^{-1} \text{ K}^{-1}$, which increases by 463% compared to that of PEG@MF. Furthermore, PEG@MF/rGO/Cu-C displays great light-to-thermal and electric-to-thermal conversion capabilities, thermal cycle stability, light-to-thermal cycle stability, and shape stability, showing promising application prospects in different aspects.

KEYWORDS

Cu-coated network, melamine foam, PEG, reduced graphene oxide, stabilized multifunctional phase change materials

1 | INTRODUCTION

Phase change materials (PCMs) are substances that can charge or discharge thermal energy through a phase transition. Due to their potential as clean and sustainable energy storage materials, PCMs have been widely used in different aspects, such as building air-conditioning, device temperature controlling, and waste heat recovery.^{1–4} In particular, organic PCMs (OPCMs), such as polyethylene glycol (PEG) and paraffin wax (PW), exhibit high phase change enthalpies, suitable phase change temperatures, low corrosivity and toxicity, and good thermal stability, which make their use over the other types of PCMs advantageous.⁵ However, the large-scale application of OPCMs has severely been limited by their low thermal conductivity and leakage above their melting point.⁶ Thus, numerous studies have been conducted for addressing these shortcomings to improve the comprehensive performance of OPCMs.

Leakage shortens the service life of OPCMs. To prevent this, leakage-proofing methods have been designed in numerous studies.^{7–10} Microencapsulation, a method where the contact of OPCMs to the external environment is completely eliminated by encapsulating them in microcapsules, has been used to prevent leakage.¹¹ Lin et al.¹² used ethyl cellulose (EC), which was reported to have good thermal stability, as a shell material for the microencapsulation of myristic acid (MA). The MA@EC composite exhibited excellent thermal cycle stability, and its melting enthalpy decreased only by 0.7% after 100 thermal cycles. Kaya et al.¹³ employed cellulose nanofibrils (CNFs) as a shell material for *n*-hexadecane (HDE) and prepared an HDE@CNF composite. After 200 thermal cycles, approximately 99% of the phase change enthalpy of the composite was retained, thereby proving its excellent thermal stability. Therein, the use of CNF was also beneficial because it did not require the use of other reagents or catalysts for the formation of microcapsules. Nevertheless, the small encapsulation ratio of PCMs, which leads to a relatively low phase change enthalpy, achieved in microencapsulation hinders their application in fields that require high energy density.¹⁴

Encapsulating OPCMs in porous supports with a high specific area and large pore volume is also an efficient strategy that confines the liquid phase through physical forces, such as surface tension, capillary effect, and hydrogen bonding.¹⁵ Atinafu et al.¹⁶ synthesized a series of micro/mesoporous metal supramolecular gels with a large specific surface area to encapsulate PEG based on the supramolecular gelation effect. The prepared composites exhibited excellent thermal cycle stability, and the best performing sample had an encapsulation ratio as high as 90%. Chen et al.¹⁷ developed a highly graphitized

three-dimensional (3D) carbon aerogel through a one-step calcination of carbon quantum dots. The aerogel was used as a framework for PEG, and the resulting shape-stabilized OPCM with a high PEG content (85%) exhibited an excellent leakage-proofing property. Yan et al.¹⁸ fabricated novel hierarchically porous hollow carbon spheres (HCSPs) through a sacrificial template method, where an SiO₂ core was dissolved using a strong alkali to build the hollow structure. The produced HCSPs were used to encapsulate 1-octadecanol (ODA), and the resulting ODA@HCSP composite had a high package ratio of 80% and a good thermal cycle stability. Moreover, the use of porous supports also provides an efficient route for the design of multifunctional PCMs as they can be easily functionalized through the incorporation of functional fillers, such as metal oxide,¹⁹ graphene,²⁰ carbon nanotubes,²¹ MXene,²² and so forth.

Over the recent years, melamine foam (MF) has gained considerable research attention due to its 3D-connected porous structure, excellent flexibility, low density, and great flame retardancy, which make it a suitable material for supercapacitors,²³ solar steam generators,²⁴ electromagnetic wave absorbers,²⁵ and so forth. However, the highly open macroporous structure of MF results in a relatively weak physical force toward OPCMs, leading to a poor encapsulation ability. Thus, the modification of MF before the impregnation of OPCMs is necessary. Xue et al.²⁶ fabricated an MF-based framework through a pyrolysis treatment of a precursor containing MF, reduced graphene oxide (rGO), and graphene nanoplatelets (GNPs). The obtained carbonized MF/rGO/GNP aerogel exhibited an excellent encapsulation ability, and 98.31% of PW was absorbed. Du et al.²⁷ introduced polydopamine (PDA) into MF and fabricated a PDA-modified MF (PMF) network. PDA contains a large number of polar hydroxyl and imine groups, which compensated for the lack of polar functional groups of MF. The PMF network was used to encapsulate PEG, and the resulting PEG@PMF composite had a PEG content as high as 98.7%. Additionally, MF can be easily altered through the incorporation of functional fillers that can confer different functions to the resulting composites, which can make it suitable for light-to-thermal conversion,²⁸ electric-to-thermal conversion,²⁹ light-activated shape memory,³⁰ and so forth. These properties of MF show its great potential as a support for OPCMs.

In this study, a novel carbonized MF/rGO/Cu network was constructed for fabricating stabilized multifunctional PCMs based on the design concept shown in Figure 1. First, MF was chosen as porous support for the functional fillers and the PCMs, providing the foundation for superior stabilities. Second, rGO was introduced into MF as a pore modifier, which can enhance the encapsulation ability of the support. Besides, rGO also served as a photon harvester,

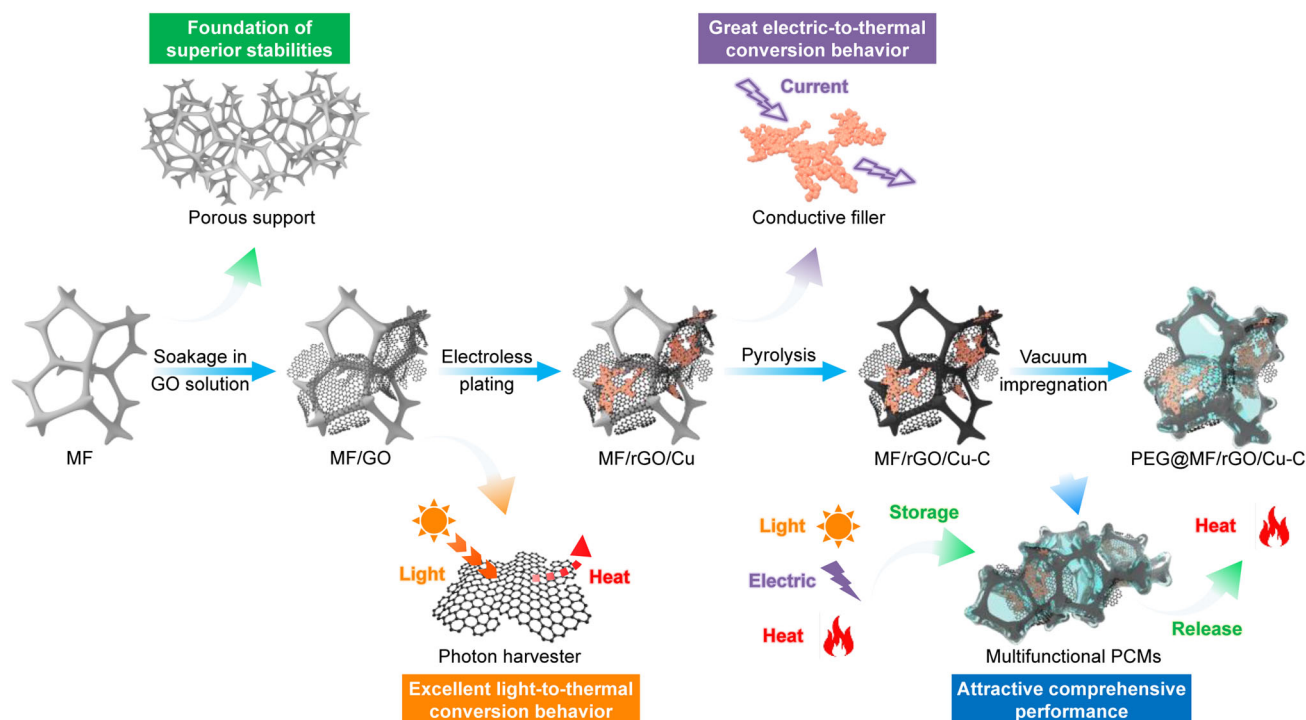


FIGURE 1 Design concept and fabrication procedure for the novel carbonized MF/rGO/Cu network and the corresponding composite PCM. MF, melamine foam; PCM, phase change material; rGO, reduced graphene oxide

which can endow the composite with excellent light-to-thermal conversion capability. Third, the Cu particles acted as conductive reinforcement, which can enhance the extremely low thermal conductivity of MF and confer great electric-to-thermal conversion ability to the composite. Thus, this study aims to produce a multifunctional PCM suitable for various applications through the outlined novel strategy.

2 | MATERIALS AND METHODS

2.1 | Materials

Commercial MF (Figure S1A,B) was obtained from BASF Co., Ltd. GO (Figure S1C,D) was prepared through Hummer's method³¹ using crystalline flake graphite from XFNANO Materials Tech Co., Ltd. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, glucose (GLC), NaOH, and $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ were obtained from Xilong Scientific Co., Ltd. Polyvinylpyrrolidone (PVP) with an average molecular weight of 58000 and PEG-6000 were purchased from Aladdin Reagent Co., Ltd.

2.2 | Synthesis of the carbonized MF/rGO/Cu (MF/rGO/Cu-C) network

Commercial MF was cut into rectangular pieces ($3\text{ cm} \times 3\text{ cm} \times 1\text{ cm}$) and then soaked in a GO solution with a

concentration of 6 mg mL^{-1} for 5 h. To obtain the MF/GO, the immersed MF pieces were removed from the GO solution and dried in a vacuum oven at 60°C . Thereafter, 12.5 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 4.95 g of GLC, and 0.6 g of PVP were dissolved in 200 mL of deionized water (DI). The MF/GO samples were then immersed into the resulting solution and maintained at 60°C for 12 h in a water bath. The following solutions were then added dropwise consecutively—a solution containing 14 g of NaOH in 80 mL of DI, and a solution composed of 55 mL of $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ and 300 mL of DI. The samples were then washed with DI several times and dried in a vacuum oven at 60°C to obtain MF/rGO/Cu. Finally, the MF/rGO/Cu was carbonized at 600°C under an N_2 atmosphere for 2 h to prepare carbonized MF/rGO/Cu network, denoted as MF/rGO/Cu-C. For comparison, networks without Cu were also fabricated by directly reducing the MF/GO with the $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ solution, followed by the same drying (MF/rGO) and carbonization (MF/rGO-C) procedures.

2.3 | Synthesis of the composite PCMs based on the MF-based networks

PEG-6000 was loaded into MF-based networks, including MF/rGO, MF/rGO-C, MF/rGO/Cu, and MF/rGO/Cu-C, through a vacuum-assisted impregnation method performed

at 80°C for 15 h in a vacuum oven. Correspondingly, the obtained composites are labeled as PEG@MF/rGO, PEG@MF/rGO-C, PEG@MF/rGO/Cu, and PEG@MF/rGO/Cu-C. For comparison, PEG@MF was also prepared by loading PEG-6000 into pure MF.

2.4 | Characterizations and measurements

Scanning electron microscopy (SEM; JSM-7610F), X-ray diffraction (XRD; SmartLab 9 kW, Cu K α), Fourier-transform infrared spectroscopy (FTIR; Nicolet 6700), X-ray photoelectron spectroscopy (XPS; ESCALAB 250xi), and Raman spectroscopy (LabRAM HR800, 533 nm) were performed to characterize the as-prepared MF-based networks and the corresponding composite PCMs. Differential scanning calorimetry (DSC; DSC 250) was performed from -10°C to 110°C at a rate of $10^{\circ}\text{C min}^{-1}$ to study the phase change properties of the composite PCMs. Thermal conductivities of the composite PCMs were measured at 25°C using a transient plane source conductometer (DZDR-S). The light-to-thermal conversion behavior of some representative samples was investigated using a xenon lamp (CEL-HXF300-T3; Beijing CEAULIGHT Technology Co., Ltd) and a multichannel data recorder (Shenzhen Kailidi Technology Co., Ltd); the intensity of the light was set at 150 mW cm^{-2} . The electric-to-thermal conversion behavior was evaluated through a platform composed of a DC power supply (HY3005MT; Hangzhou HYELEC Co., Ltd) and a multichannel data recorder, and a constant voltage of 2 V was applied.

3 | RESULTS AND DISCUSSION

3.1 | Characterizations of the MF-based networks

Before the deposition of a coating, the surface of a substrate is typically modified using strong acid and alkali to increase its roughness and improve its wetting behavior for obtaining a high-quality deposit.^{32–34} To preserve the network structure, MF was not modified using a highly acidic or alkaline solution due to its poor stability under such harsh conditions. Herein, the introduction of GO into MF can improve the hydrophilicity of the network. Moreover, instead of methods conducted in strong alkaline solution, a glucose pre-reduction method with relatively mild conditions was employed for the deposition of Cu. The reaction processes involved are shown in Equations (1–3).³⁵

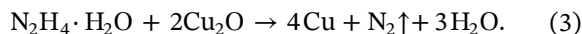
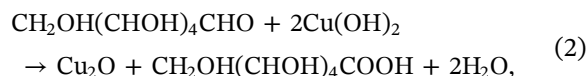
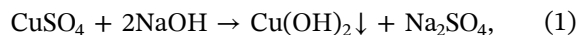


Figure 2 shows the SEM images of the MF-based networks, namely MF/GO, MF/rGO/Cu, and MF/rGO/Cu-C. All the samples exhibited a porous netted structure due to the template effect of MF. In the SEM images of MF/GO (Figure 2A–C), the adhesion of GO on the MF support was clearly observed. Figure 2D–F shows the SEM images of the MF/rGO/Cu network. Unlike the MF support, the graphene sheets were densely covered with Cu particles, which is possibly due to the difference between the hydrophilicity of GO and MF. MF/rGO/Cu-C (Figure 2G–I) exhibited a more 3D structure after carbonization, wherein the MF network can hardly be seen at low magnification. Under high magnification, it can be observed that the combination between rGO and MF support became closer compared with MF/rGO/Cu. In addition, morphologies of networks without Cu were also investigated, and the SEM images are illustrated in Figure S2. MF/rGO (Figure S2A–C) showed a similar morphology to MF/GO. After being carbonized, the rGO sheets in MF/rGO-C (Figure S2D–F) became curly, which exhibited a different morphology from the graphene sheets in MF/rGO/Cu-C. This indicates that the deposition of Cu particles possibly improved the stiffness of the graphene sheets, contributing to the more 3D structure of the MF/rGO/Cu-C.

Figure 3A shows the XRD patterns of the MF-based networks. A diffraction peak at $2\theta = 10^{\circ}$, which can be indexed to the GO absorbed on the MF support, was observed in the pattern of the MF/GO.³⁶ Characteristic peaks at $2\theta = 43.5^{\circ}$, 50.5° , and 74.3° ,³⁷ corresponding to Cu, were observed in the pattern of MF/rGO/Cu. This indicates the successful loading of Cu particles into the MF/rGO network, which corresponds to the energy dispersive X-ray spectroscopy (EDS) elemental maps of MF/rGO/Cu shown in Figure S3. Meanwhile, the peak of GO was found to disappear, attributing to the reduction of most functional groups on GO. Then, a peak at $2\theta = 26.3^{\circ}$ that can be associated with rGO was observed in the pattern of MF/rGO/Cu-C, indicating the degradation of the remaining oxygen-containing groups of graphene sheets after carbonization.³⁸ Similar phenomenon can be observed in the patterns of MF/rGO and MF/rGO-C, indicating a similar evolution procedure of graphene sheets absorbed in the network.

Figure 3B shows the FTIR spectra of the MF-based networks. In the spectrum of MF/GO, the peaks located at

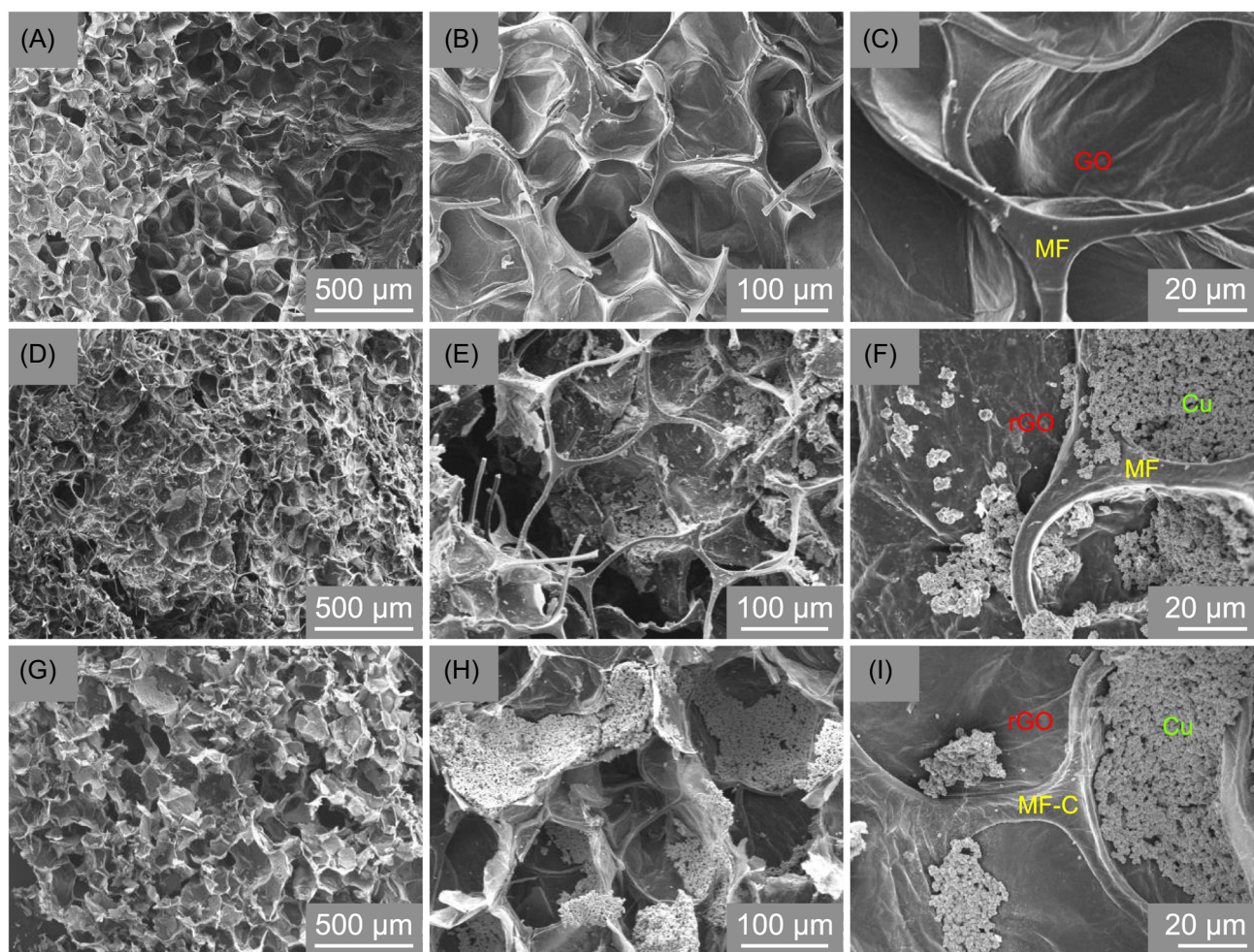


FIGURE 2 SEM images of (A–C) MF/GO, (D–F) MF/rGO/Cu, and (G–I) MF/rGO/Cu-C. MF, melamine foam; rGO, reduced graphene oxide; SEM, scanning electron microscopy

3428.6, 1635.1, and 808.1 cm^{-1} are related to the superposition of N–H and O–H stretching, the accumulation of C=N and C=O stretching, and the triazine ring bending vibration, respectively,^{28,39,40} demonstrating the successful adsorption of GO in MF. For MF/rGO, the intensities of the peaks at 3428.6 and 1635.1 cm^{-1} decreased, which implies that most of the oxygen-containing groups on GO were reduced. After carbonization (MF/rGO-C), the intensities of these two peaks decreased further, and the peak at 808.1 cm^{-1} was found to disappear. These observations confirm the carbonization of the MF framework as well as the further pyrolysis of the remaining oxygen-containing groups. Additionally, the spectra of MF/rGO/Cu and MF/rGO/Cu-C were similar to those of the networks without Cu. This indicates that the deposition of Cu had little effect on the chemical structure of the samples. Besides, XPS (Figure 3C–F) was also conducted to confirm the chemical state of the networks, and its spectra shows great correspondence to the XRD and FTIR results, where the successful

incorporation of Cu and the degradation of most of the oxygen-containing groups were confirmed.^{41,42}

Figure 4A shows the Raman spectra of the MF-based networks. All networks exhibited two main peaks, including the D-band at 1340 cm^{-1} corresponding to amorphous carbon and the G-band at 1596 cm^{-1} associated with the sp^2 -hybridization of graphitized carbon.⁴³ To calculate the integrated intensity ratio of these two bands (I_D/I_G), a peak fitting process was performed, and the calculated results are shown in Figure 4B–F. MF/GO (Figure 4D) showed the highest I_D/I_G ratio of 2.260 due to massive surface defects related to various oxygen-containing groups. The value of I_D/I_G decreased significantly after MF/GO was reduced (Figure 4B) or coated with Cu (Figure 4C). This demonstrates the successful reduction of most of the oxygen-containing groups. Lastly, the I_D/I_G ratios of MF/rGO-C (Figure 4E) and MF/rGO/Cu-C (Figure 4F) were slightly lower than those of their uncarbonized counterparts, which confirms further degradation of the remaining oxygen-containing groups during carbonization.

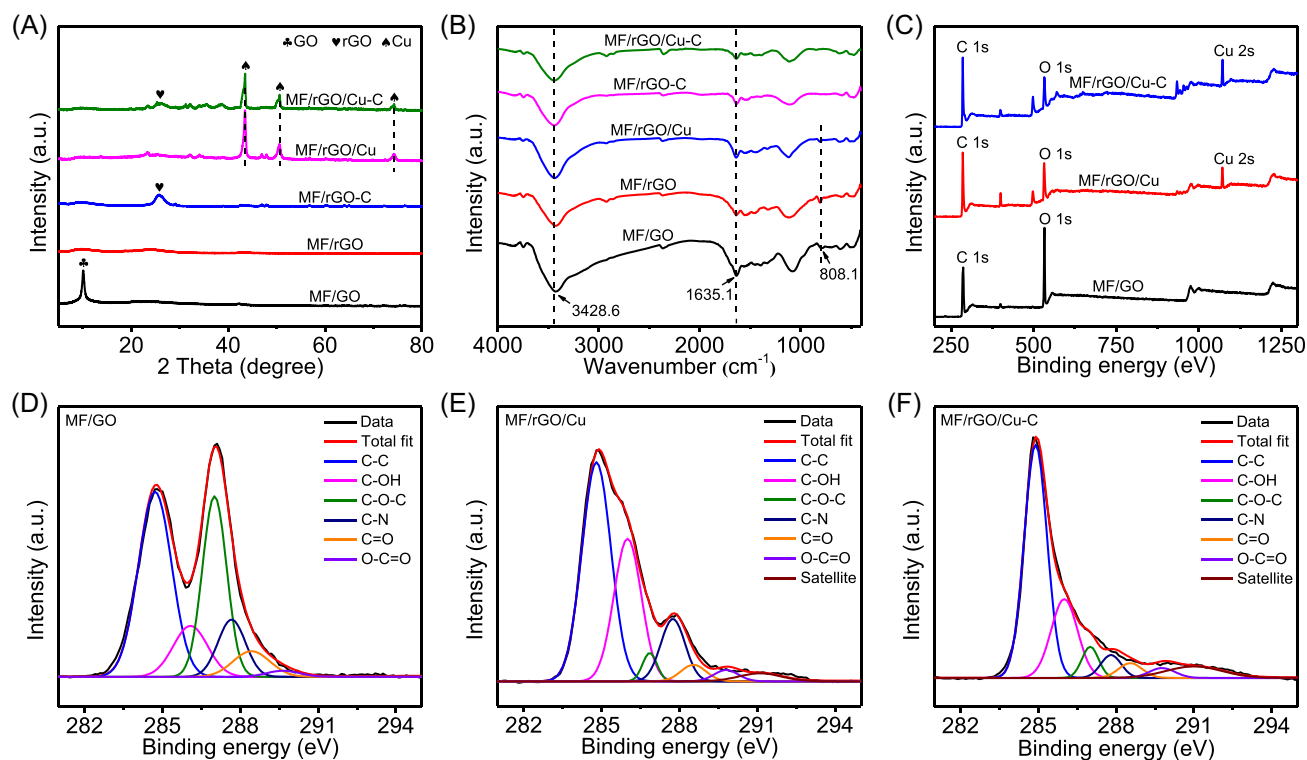


FIGURE 3 (A) XRD patterns and (B) FTIR spectra of the MF-based networks; (C) XPS survey spectra of MF/GO, MF/rGO/Cu, and MF/rGO/Cu-C, and C 1s spectra of (D) MF/GO, (E) MF/rGO/Cu, and (F) MF/rGO/Cu-C. FTIR, Fourier-transform infrared; MF, melamine foam; rGO, reduced graphene oxide; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction

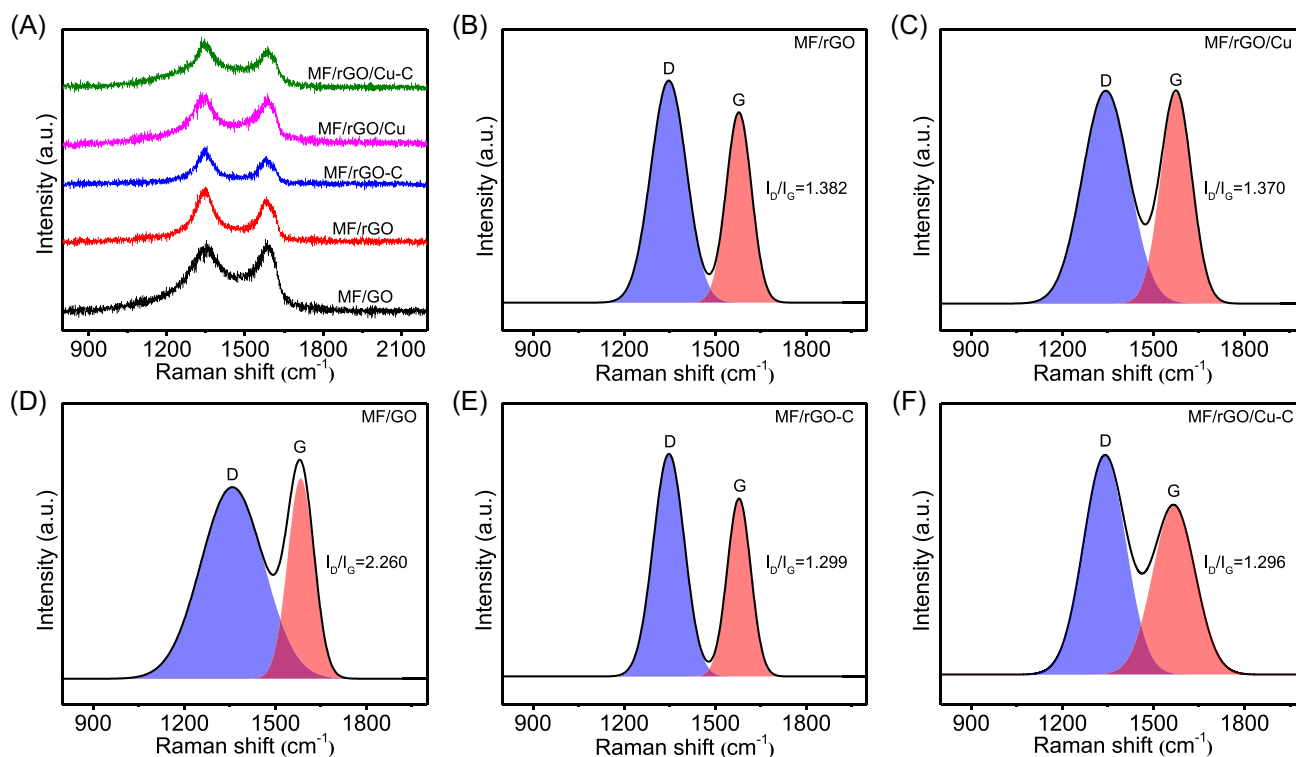


FIGURE 4 (A) Raman spectra and (B-F) peak fitting results of the MF-based networks. MF, melamine foam

3.2 | Encapsulation ability of the MF-based networks

Figure S4 shows the maximum loading determination experiment performed to evaluate the encapsulation ability of the as-prepared MF-based networks, and the results are summarized in Table 1. MF exhibited continuous leakage even after 96 h, which indicates its poor encapsulation ability, which can be attributed to the highly open macropore structure of the foam. The introduced rGO sheets modified the macropore structure of the MF framework, which led to short leakage time and high maximum loadings observed for MF/rGO and MF/rGO/Cu. Moreover, carbonized samples exhibited shorter leakage time and higher maximum loadings probably due to the twisting of rGO sheets in the

TABLE 1 Leakage time and maximum loading of the MF-based networks

Sample	Leakage time (h)	Maximum loading (%)
MF	>96	–
MF/rGO	26	87.0
MF/rGO-C	18	91.6
MF/rGO/Cu	16	84.3
MF/rGO/Cu-C	12	86.1

Abbreviations: MF, melamine foam; rGO, reduced graphene oxide.

MF/rGO-C network or the optimized stack of rGO/Cu platelets in the MF/rGO/Cu-C, both of which further improved the pore structure of the networks.

3.3 | Characterizations of the composite PCMs based on the MF-based networks

Figure 5 shows the SEM images of the composite PCMs prepared with the MF-based networks. In the PEG@MF (Figure 5A,D), the MF framework can be observed clearly, and PEG was found to cover the skeleton or fill the macropores of the MF support. On the other hand, the composites fabricated using rGO-containing networks, including PEG@MF/rGO (Figure S5A,C), PEG@MF/rGO-C (Figure S5B,D), PEG@MF/rGO/Cu (Figure 5B,E), and PEG@MF/rGO/Cu-C (Figure 5C,F), exhibited a fuller distribution of PEG, which indicates the better impregnation of PEG into the supports. This observation corresponds well to the results of the maximum loading determination experiment, which showed a large enhancement in the encapsulation abilities of the MF-based networks after the introduction of rGO into the MF support.

XRD (Figure 6A) and FTIR (Figure 6B) were performed to further confirm the compatibility between PEG and the prepared MF-based networks. The characteristic peaks of PEG at $2\theta = 19.2^\circ$ and 23.4° can be observed in the XRD patterns of all the samples, which

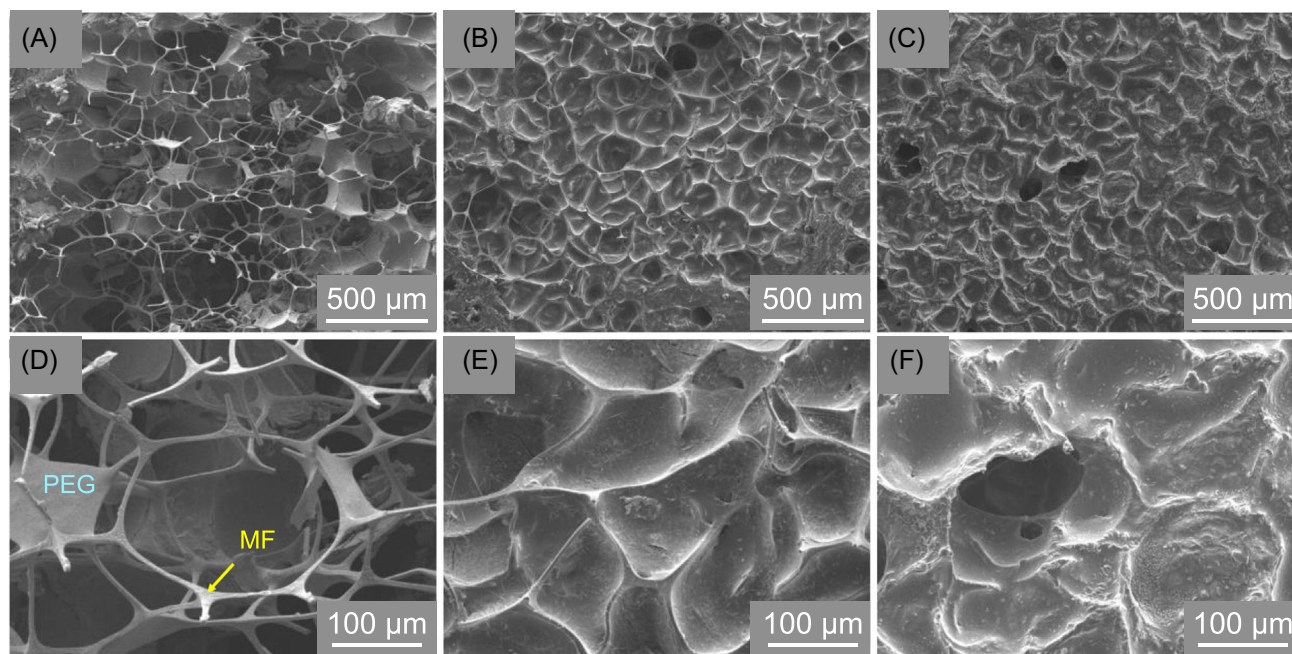


FIGURE 5 SEM images of (A,D) PEG@MF, (B,E) PEG@MF/rGO/Cu, and (C,F) PEG@MF/rGO/Cu-C. MF, melamine foam; PEG, polyethylene glycol; rGO, reduced graphene oxide; SEM, scanning electron microscopy

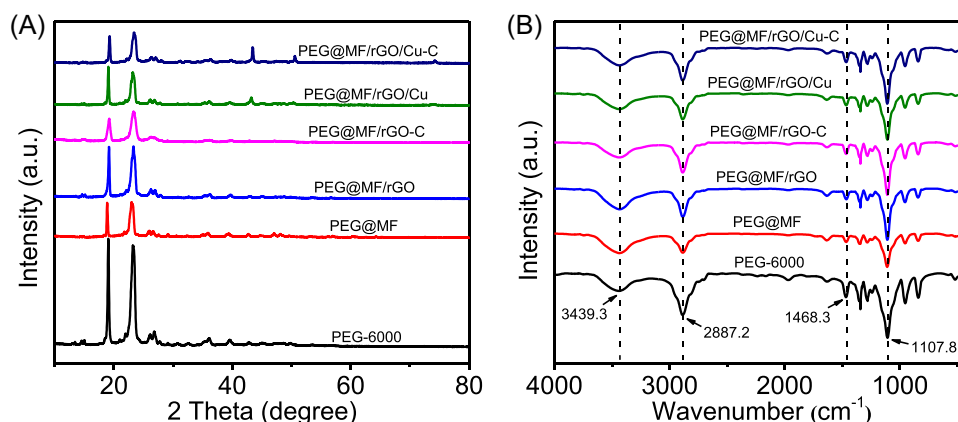


FIGURE 6 (A) XRD patterns and (B) FTIR spectra of the composite PCMs based on the MF-based networks. FTIR, Fourier-transform infrared; MF, melamine foam; PCMs, phase change materials; XRD, X-ray diffraction

indicates that the introduction of the MF-based networks did not have a considerable effect on the crystal structure of the PEG absorbed inside.⁴⁴ In Figure 6B, all the composites exhibited an FTIR spectrum similar to that of pure PEG. Characteristic adsorption peaks at 3439.3, 2887.2, 1468.3, and 1107.8 cm^{-1} , which correspond to O–H stretching, C–H stretching, C–H bending, and C–O stretching, respectively, were observed.⁴⁵ Moreover, no peaks corresponding to other phases or chemical bonds were observed in the XRD patterns and FTIR spectra, which demonstrates the physical combination and good chemical compatibility between PEG and the MF-based networks.

3.4 | Thermophysical properties of the composite PCMs based on the MF-based networks

For PCMs, the phase change temperatures and enthalpies are important parameters that directly reflect their energy storage capacities.¹⁶ Figure 7A,B show the results of the DSC analysis carried out to measure the phase change properties of the PCMs. Table 2 summarizes the corresponding thermal parameters, including the melting temperature (T_m), melting enthalpy (ΔH_m), crystallization temperature (T_c), and crystallization enthalpy (ΔH_c). All the composites exhibited similar melting and crystallization peaks to pure PEG, which indicates the maintenance of the crystalline behavior of PEG absorbed in supports. Compared to the sharp peaks observed in the DSC curves of pure PEG, broad peaks were observed in those of the composites, possibly due to the asynchronous phase change of PEG absorbed in pores of different sizes.²⁶ The T_m of the composites, including PEG@MF/rGO, PEG@MF/

rGO-C, PEG@MF/rGO/Cu, and PEG@MF/rGO/Cu-C, was slightly higher than that of pure PEG. This is possibly due to the strong interaction between PEG and the supports, which slightly delays the melting procedure of PEG.³⁷ On the other hand, PEG@MF showed a lower T_m than pure PEG, which indicates that the pristine MF may have an unignorable effect on the crystal quality of the absorbed PEG. This can also be observed in the crystallization procedure of PEG@MF, where T_c had a 5.6°C decrease. This can be attributed to the highly open macroporous structure of MF, which provides little interaction with the PCMs.

Among the composites, PEG@MF showed the lowest phase change enthalpies of 138.3 and 135.7 J g^{-1} due to the poor encapsulation ability of MF, which was confirmed by the observed continuous leakage even after 96 h of heating in the maximum loading experiments. On the other hand, PEG@MF/rGO, PEG@MF/rGO-C, PEG@MF/rGO/Cu, and PEG@MF/rGO/Cu-C registered higher phase change enthalpies than PEG@MF, which indicates a considerable enhancement of the encapsulation ability after the modification of the MF support. Furthermore, the energy storage capability (φ) of the composite PCMs was calculated using Equation (4):

$$\varphi = \frac{\Delta H_{m/\text{comp}} + \Delta H_{c/\text{comp}}}{R(\Delta H_{m/\text{PEG}} + \Delta H_{c/\text{PEG}})} \times 100\%, \quad (4)$$

where R , $\Delta H_{m/\text{comp}}$, and $\Delta H_{c/\text{comp}}$ represent the maximum loading (Table 1), melting enthalpy, and crystallization enthalpy of the composite PCMs, respectively; $\Delta H_{m/\text{PEG}}$ and $\Delta H_{c/\text{PEG}}$ correspond to the phase change enthalpies of pure PEG.¹⁶ The values of φ for all the composites, except for the PEG@MF which has no value for the maximum loading, were >90% (Table 2). This demonstrates that the composite PCMs fabricated with

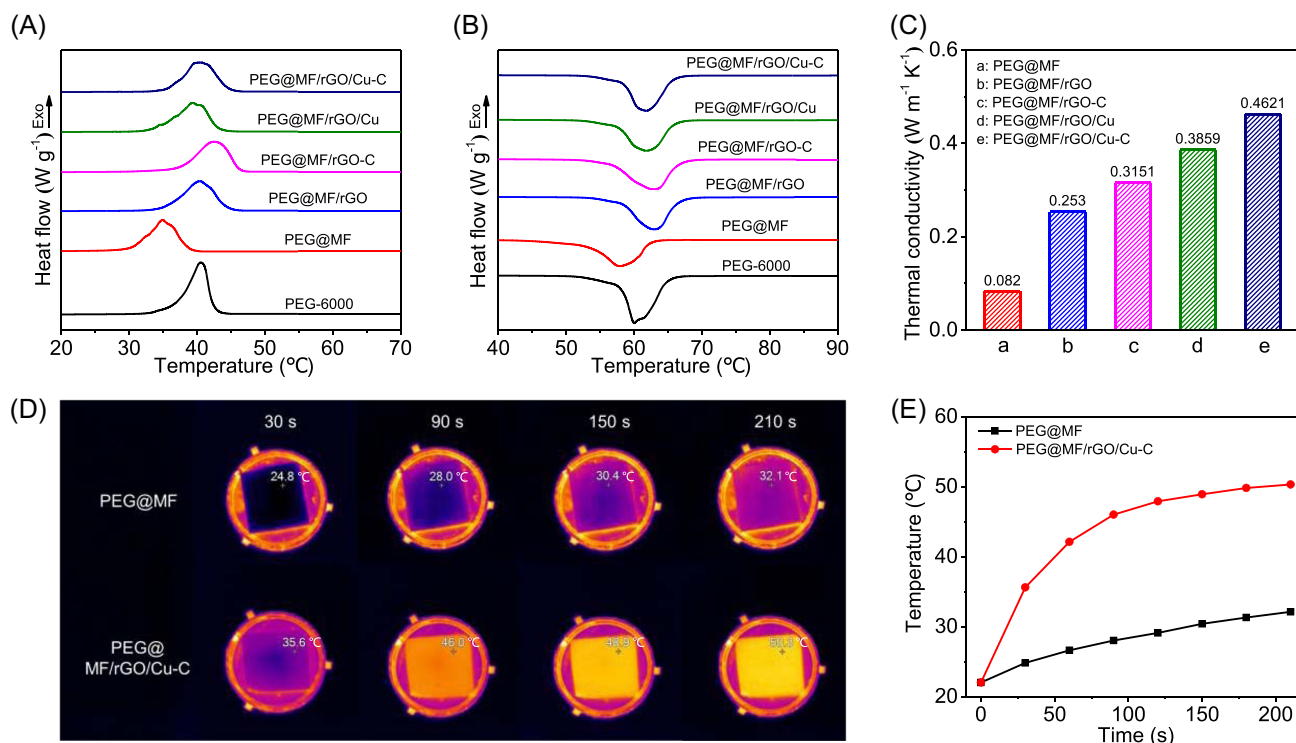


FIGURE 7 (A,B) DSC curves and (C) thermal conductivities of the PCMs; (D,E) thermal response behavior of PEG@MF and PEG@MF/rGO/Cu-C. DSC, differential scanning calorimetry; MF, melamine foam; PCMs, phase change materials; PEG, polyethylene glycol; rGO, reduced graphene oxide

TABLE 2 Thermal parameters of the composite PCMs based on the MF-based networks.

Sample	T_m (°C)	ΔH_m (J g ⁻¹)	T_c (°C)	ΔH_c (J g ⁻¹)	φ (%)	F_c (%)
PEG-6000	60.0	185.3	40.5	180.9	100	100
PEG@MF	57.9	138.3	34.9	135.7	–	–
PEG@MF/rGO	63.0	148.6	40.4	146.4	92.6	92.2
PEG@MF/rGO-C	62.9	161.3	42.5	157.6	95.1	95.0
PEG@MF/rGO/Cu	61.8	148.4	39.3	142.7	94.3	95.0
PEG@MF/rGO/Cu-C	61.8	148.3	40.2	143.9	92.7	92.9

Abbreviations: MF, melamine foam; PEG, polyethylene glycol; rGO, reduced graphene oxide.

prepared networks are promising candidates for energy storage applications.

In addition, to further investigate the effect of the prepared networks on the crystalline behavior of PEG, the relative crystallinity (F_c) of the composite PCMs was also calculated using Equation (5).

$$F_c = \frac{\Delta H_{m/comp}}{R \times \Delta H_{m/PEG}} \times 100\%. \quad (5)$$

As illustrated in Table 2, the F_c values of the composites, including PEG@MF/rGO, PEG@MF/rGO-C, PEG@MF/

rGO/Cu, and PEG@MF/rGO/Cu-C, were >90%. This indicates that the introduction of the prepared networks had little effect on the crystalline behavior of PEG.

Besides enthalpy, thermal conductivity is also an important thermophysical parameter, which indicates the heat transfer rate during the energy storage process. Figure 7C summarizes the thermal conductivities of the composites prepared using different MF-based networks. PEG@MF exhibited an extremely low thermal conductivity of 0.0820 W m⁻¹ K⁻¹ because of the high thermal resistance of MF. The introduced graphene sheets on the MF support supply a heat transfer path, and thus

PEG@MF/rGO exhibited an improved thermal conductivity of $0.2530 \text{ W m}^{-1} \text{ K}^{-1}$. Cu particles were also introduced into the network as thermally conductive reinforcer considering the superior heat transfer properties of metals. PEG@MF/rGO/Cu exhibited further improvement in the thermal conductivity from 0.2530 to $0.3859 \text{ W m}^{-1} \text{ K}^{-1}$ due to the introduction of Cu. Overall, PEG@MF/rGO/Cu-C registered the highest thermal conductivity of $0.4621 \text{ W m}^{-1} \text{ K}^{-1}$, which increased by 463% compared with that of PEG@MF. This is probably due to the formation of the 3D-connected thermal conductive network after carbonization.

PEG@MF and PEG@MF/rGO/Cu-C samples with the same thickness were placed on a hot stage held at 55°C to further investigate the thermal response rate during the heating process. Temperatures of the top surfaces of the samples were recorded using an infrared camera. As shown in Figure 7D,E, PEG@MF/rGO/Cu-C exhibited a much faster response rate than PEG@MF. Within 210 s, the temperature of PEG@MF/rGO/Cu-C rapidly increased to 50.3°C , while PEG@MF only reached 32.1°C . This observation can be attributed to the difference between the thermal conductivity of the samples, corresponding well to the results of the thermal conductivity measurements.

3.5 | Energy conversion properties of PEG@MF/rGO/Cu-C

Besides the direct storage of thermal energy, the indirect storage through energy conversion processes can increase the suitability of PCMs in a wide variety of applications. Light-to-thermal conversion behaviors of pure PEG and PEG@MF/rGO/Cu-C were investigated using the platform shown in Figure 8A, and the results are illustrated in Figure 8C. A rapid temperature increase to 73.0°C , which is 13°C higher than the melting point of PEG, was achieved within 240 s of irradiation of PEG@MF/rGO/Cu-C. Within the same time, the temperature of pure PEG only reached 48.9°C . The light-to-thermal conversion efficiency (η_l) of the samples was calculated using Equation (6):

$$\eta_l = \frac{m \times \Delta H_m}{P \times S \times (t_f - t_s)} \times 100\%, \quad (6)$$

where m and S represent the mass and surface area of the sample, respectively; P refers to the intensity of the light; t_s and t_f correspond to the start and finish times of the light-driven melting process, respectively.⁴⁶ The calculated η_l for PEG@MF/rGO/Cu-C was as high as 92.7%, which indicates

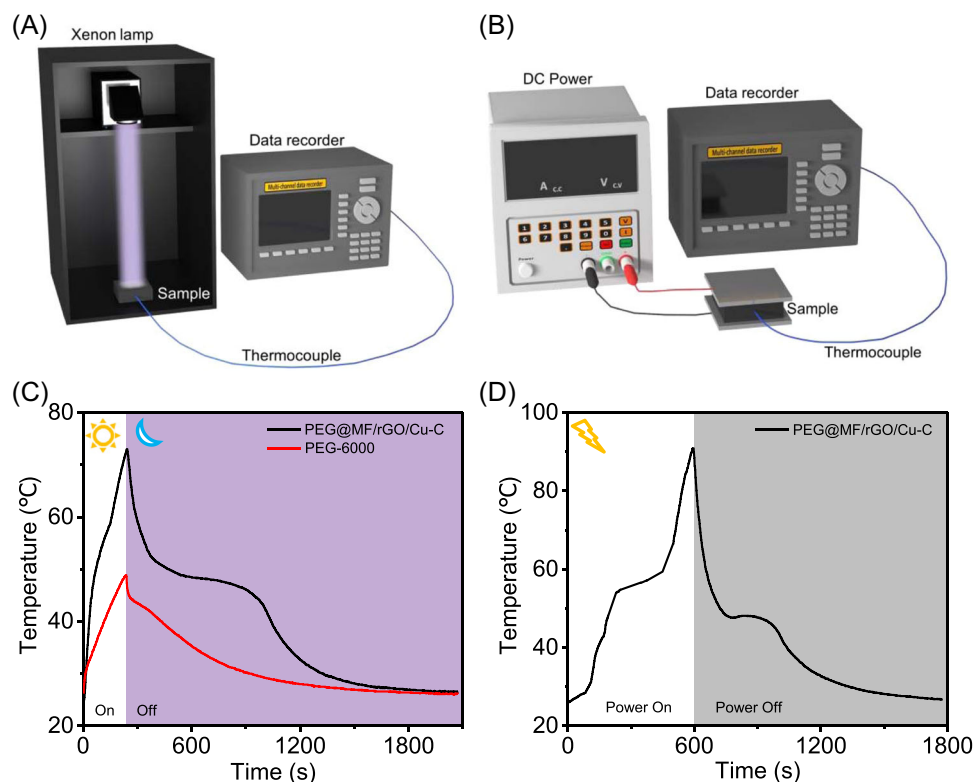


FIGURE 8 (A) Light-to-thermal and (B) electric-to-thermal conversion experiment platforms; (C) light-to-thermal conversion behavior of PEG-6000 and PEG@MF/rGO/Cu-C; (D) electric-to-thermal conversion behavior of PEG@MF/rGO/Cu-C. MF, melamine foam; PEG, polyethylene glycol; rGO, reduced graphene oxide

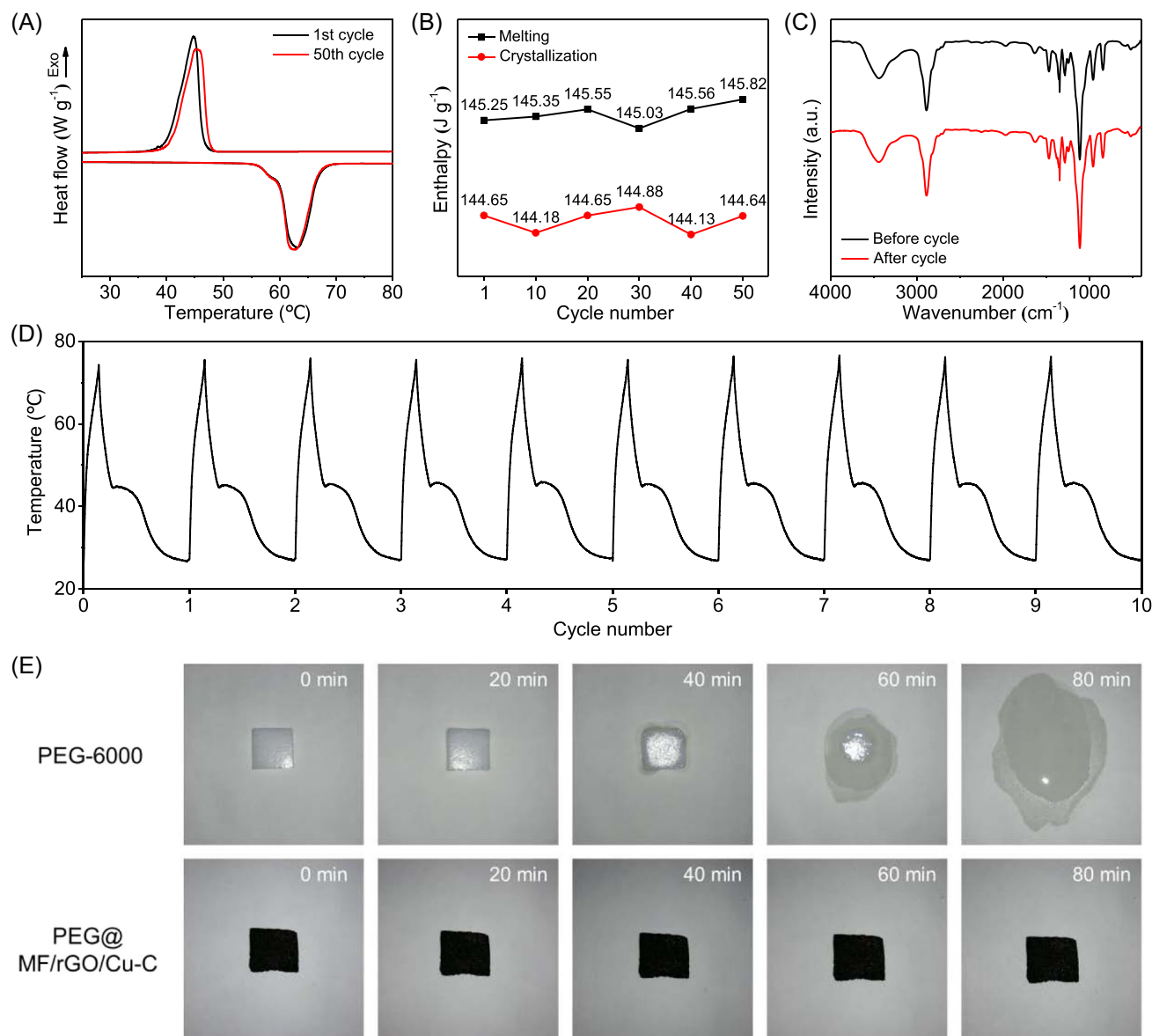


FIGURE 9 (A–C) Thermal and (D) light-to-thermal cycle behavior of PEG@MF/rGO/Cu-C; (E) leakage-proofing test of PEG-6000 and PEG@MF/rGO/Cu-C. MF, melamine foam; PEG, polyethylene glycol; rGO, reduced graphene oxide

its excellent light-to-thermal conversion behavior that can be attributed to the great photon harvesting property of rGO. The temperatures of the samples began to decrease after the xenon lamp was switched off (Figure 8C). A plateau corresponding to a phase change during cooling can not be observed in the cooling curve of pure PEG. On the other hand, the cooling curve of PEG@MF/rGO/Cu-C displayed a crystalline plateau that lasted for approximately 10 min. This demonstrates the nearly complete melting of the absorbed PEG during irradiation owing to the synergistic effect of the excellent light-to-thermal property and the enhanced thermal conductivity.

In addition, the electric-to-thermal conversion behavior of PEG@MF/rGO/Cu-C was studied through the

setup shown in Figure 8B. On the one hand, PEG displays no electric-to-thermal conversion property due to its poor electrically conductive property; on the other, the composite exhibited a typical temperature-time curve, as depicted in Figure 8D. During the switch-on stage, the temperature of the composite increased to 90.0°C within 600 s. This can be attributed to the generation of quantities of Joule heat, which indicates the improvement of the electrically conductive property of PEG. Moreover, plateaus corresponding to melting and crystallization processes can be observed during the switch-on and switch-off stages, respectively, which demonstrates that the generated Joule heat can be stored and released by the composite. Similarly, the

electric-to-thermal conversion efficiency (η_e) was calculated using Equation (7):

$$\eta_e = \frac{m \times \Delta H_m}{U \times I \times (t_f - t_s)} \times 100\%, \quad (7)$$

where U and I refer to the voltage and current of the DC power supply, respectively.²⁹ The calculated η_e was 77.2%, which demonstrates that the electrically conductive MF/rGO/Cu-C network confers great electric-to-thermal property to the composite.

3.6 | Thermal cycle, light-to-thermal cycle, and shape stability of PEG@MF/rGO/Cu-C

Service life, which is strongly associated with stability, is a significant parameter for large-scale applications. Herein, the thermal cycle, light-to-thermal cycle, and shape stability of PEG@MF/rGO/Cu-C were studied. A thermal cycle measurement of 50 times was conducted using DSC and the results are shown in Figure 9A. The curve of the 50th cycle exhibited a good contract ratio to that of the first cycle. Figure 9B summarizes the enthalpies obtained after every 10 thermal cycles. The enthalpies of melting and crystallization obtained after 50 thermal cycles (145.82 and 144.64 J g⁻¹) did not change considerably compared to the initial values (145.25 and 144.65 J g⁻¹). Additionally, the FTIR spectra (Figure 9C) obtained before and after the thermal cycle measurement showed great similarity, which indicates the excellent thermal cycle stability of the PEG@MF/rGO/Cu-C. Moreover, light-to-thermal cycle measurement was carried out using the setup shown in Figure 8A, and the obtained pattern is illustrated in Figure 9D. The curves obtained after every cycle were highly similar to that of the first cycle, demonstrating that the composite also has excellent light-to-thermal cycle stability. Finally, the shape stability of pure PEG and PEG@MF/rGO/Cu-C was evaluated by a leakage-proofing test, where samples were placed on a piece of filter paper and then heated at 80°C. Figure 9E shows the digital photos of the samples taken every 10 min until PEG was completely melted. After heating for 80 min, the pure PEG sample completely melted, but no leakage was observed for the PEG@MF/rGO/Cu-C composite. This indicates the superior shape stability and leakage-proofing property of PEG@MF/rGO/Cu-C, which can be attributed to the good absorption of the melted PEG in the pores of the MF/rGO/Cu-C network during heating. Thus, the excellent encapsulation ability of MF/rGO/Cu-C provides great potential for the large-scale application of the prepared PEG@MF/rGO/Cu-C composite PCM.

4 | CONCLUSION

An MF/rGO/Cu-C framework for fabricating stabilized multifunctional PCMs was synthesized through the carbonization of a network consisting of MF, rGO, and Cu particles. PEG was selected as a thermal energy storage material and was impregnated into the framework through a vacuum-assisted impregnation method. The obtained PEG@MF/rGO/Cu-C composite showed a high maximum loading (86.1%) without leakage. This revealed the great encapsulation ability of the framework, which was attributed to the optimization of the pore structure of MF brought by the introduction of rGO and the carbonization process. The phase change enthalpies for melting and crystallization of the PEG@MF/rGO/Cu-C composite were 148.3 and 143.9 J g⁻¹, respectively. The calculated energy storage capability was 92.7%, which demonstrated that the introduction of the MF/rGO/Cu-C framework did not have a considerable effect on the crystalline behavior of PEG. Due to the 3D-connected thermally conductive network composed of carbonized MF, rGO, and Cu particles, the composite had an enhanced thermal conductivity equal to 0.4621 W m⁻¹ K⁻¹, which was 463% higher than that of PEG@MF. The composite also displayed great light-to-thermal and electric-to-thermal conversion capabilities owing to the synergistic effect of the introduction of rGO and Cu, where rGO served as a photon harvester and Cu acted as an electrically conductive reinforcer. Moreover, the framework also conferred excellent thermal cycle, light-to-thermal cycle, and shape stability to the PEG@MF/rGO/Cu-C composite, which further demonstrated the excellent encapsulation ability of MF/rGO/Cu-C. Thus, this study provided a promising strategy for the preparation of stabilized multifunctional PCMs that show great potential for various applications.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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